

PROJECT PRESENTATION

A 3D endless runner built with Unity 6

Game project · 11/2026

PRESENTATION OUTLINE

Five parts of this presentation

1. Project introduction: what the game is, how it plays, what has been built
2. Story: the setting, how a run unfolds and the tone
3. Technologies applied: tools, Unity packages and techniques in code
4. Process: phases, teamwork and source control
5. The problem: goals, constraints, challenges and how we solved them

Run as far as you can. Never stop.

3

lanes, the character runs automatically through the city

4

actions: left, right, jump, slide

Endless

no final level

- A third-person 3D endless runner, built with Unity 6
- Change lanes, jump and slide to dodge obstacles and collect coins and power-ups
- Success is measured by distance, coins and score
- Reference game: Subway Surfers

Project information

Item	Details
Game title	Escape Dash
Genre	3D Endless Runner
Players	Single player, one character per run
Platform	PC, keyboard controls
Engine, language	Unity 6 (6000.3), C#, Universal Render Pipeline
Camera	3D, fixed perspective camera behind and above the character
UI language	Mostly English (menu, pause, revive); a few screens are still in Vietnamese
Screens	Intro (loading), Menu and gameplay
Reference	Subway Surfers

Goals and target players

Product goals

- A complete endless runner, from the menu to Game Over and back
- A clear loop: run, dodge, collect, score, lose, replay
- Layers that bring players back: records, coins, revive, leaderboard

Experience goals

- Learn the controls in seconds, with no long tutorial
- Every loss is understood
- Short runs, so you want to retry

Target players

- Casual players who like short games, a few minutes at a time
- No prior gaming experience needed
- Playing on a computer with a keyboard

Motivation

- Beat your own record
- Climb the leaderboard
- Collect more coins
- Try harder stretches of road

Five distinctive features

Directional collisions	A head-on hit loses instantly; a side graze or a top touch only makes you stumble
4-second danger zone	After a stumble, stumbling again within 4 seconds loses
Revive with coins	Coins are both score and an asset kept across runs
Online leaderboard	Anonymous sign-in, no account needed
Five-level random road	Difficulty alternates, and there is always a lane you can dodge into

A short loop, easy to replay

1 · Observe
Look at the row of obstacles and find the empty lane

2 · Choose
Change lane, jump or slide

3 · Dodge and collect
Dodge obstacles, collect coins on the safe lane and power-ups

4 · Score
Score, distance and coins rise instantly, visible on the HUD

5 · Lose, replay
Revive with coins, or replay right away with a shortcut

Coins accumulate across runs; personal records and the leaderboard keep players coming back.

Five things the game holds to

Readable	Danger is understood at a glance: three clear lanes, every row of obstacles leaves one empty lane, three easy-to-tell obstacle types
Easy to replay	Short runs: a one-step revive, a replay shortcut, a compact Game Over screen
Instant feedback	Every action has sound and motion: jump, roll, stumble, fall, coin, collision
Rising tension, still fair	It gets harder but never creates an undodgeable situation: five difficulty levels, a safe lane, a 4-second danger zone
Personal progress	Records and a leaderboard; the road is random, so you must read it instead of memorizing it

Four decision rules

1 · Easy to read

If a change makes a situation hard to recognize, drop it.

2 · Always dodgeable

If a change creates a situation that can't be dodged, drop it.

3 · Keep it simple

If a feature needs a long explanation, simplify it or drop it.

4 · Instant reward

Every reward must be visible right away: sound, numbers on screen.

What has been built

Character

- Three-lane controls, jump and slide
- Six animation states
- A fixed Cinemachine camera

Road

- An endless road built from segments
- Five random difficulty levels
- A 120-second day-night cycle

Items

- Coins and three obstacles
- Three power-ups: Magnet, Double Score, Super Jump

Scoring

- Score = distance + 10 × coins
- Record saved on the device
- A coin wallet kept across runs

Revive and leaderboard

- Revive for 50 coins, 5-second countdown
- Online leaderboard, top 20

Interface and audio

- Intro, Menu, HUD, pause, Game Over
- Music and effects, three volume channels

The character stays still; the road runs toward them

3
lanes: left, middle, right

12
units per second when changing lanes

0.17 s
to change one lane

- The character only shifts sideways (changing lanes) and vertically (jumping); the whole road is pulled back toward the character
- Speeding up just means pulling the road faster; the character never drifts far from the origin
- Lane positions on the x axis: −2, 0, +2 units; the middle lane is the starting lane
- Capsule collider 1.8 tall with radius 0.35; a kinematic Rigidbody, identified by the Player tag

Four actions, learned in seconds

A / D or ← / →	Shifts exactly one lane per press, never past an edge lane; pressing both at once counts as left
W / Space / ↑	Jump: only possible on the ground; jumping cancels a slide in progress
S / ↓	Slide: can't be pressed again mid-slide; press in the air to dive down
Esc	Pause and open the volume settings; Esc in settings goes back to pause

Once you have lost, the character takes no more input. The game is played entirely on the keyboard.

Six animation states

State	When	Notes
Run	Normal running; after landing, ending a slide, ending a stumble, reviving	Default
JumpStart	Just left the ground	—
JumpLoop	Hovering in the air	Starts after 0.25 s, not entered while sliding
Roll	Sliding	—
Hit	Just stumbled on an obstacle	0.6 s, only when grounded and not sliding
Death	Lost	—

States blend over 0.1 s so the motion never looks jerky.

Coins: both score and an asset

+10

points per coin

+1

coin into the wallet, kept across runs

3

coins per row, 1.5 units apart

- Placed on the row's safe lane, so collecting coins never forces you into the danger zone
- Spins 180 degrees per second, with a sound when collected
- Collected coins come back when the road segment is reused
- At most 24 coins per segment (a pre-built buffer); a segment has 3 rows, so at most 9 are used

Three power-ups

Magnet · 10 s

- Item: a magnet bag
- Pulls every coin within a radius of 8 toward the player at 28 units per second
- Coins closer than 0.4 are picked up automatically
- A radius of 8 covers all three lanes and about one row ahead

Double Score · 10 s

- Item: a star
- Doubles the distance points and the value of each coin, both score and wallet
- Only counts for road covered while it is active

Super Jump · 8 s

- Item: a pair of shoes
- Multiplies jump velocity by 1.8: about 4.7 high (3.2× normal) and about 1.31 s in the air
- Sliding in mid-air dives down proportionally faster

Power-up rules

Quantity
At most one power-up per segment, replacing that row's coins; two power-ups are at least 4 segments apart

Weighting
All three types are equally likely (2:2:2). The chance of a power-up is 100% at level 1 and 25% at levels 2 to 5

Picking up again
The timer restarts from the beginning; it does not stack

Concurrent
Power-ups of different types run at the same time, independently

On losing
All are cleared; reviving does not restore them

On the HUD
An icon with a countdown ring and the seconds left, flashing during the last 3 seconds

From menu to Game Over



Intro and menu

- Intro: a LOADING bar, at least 2 seconds, then the menu
- Menu: START, SETTINGS, LEADERBOARD, QUIT
- SETTINGS: three volume sliders and a BACK button

During play

- HUD: score, coins, distance, speed, power-up icons
- Esc opens PAUSE GAME: RESUME, RESTART, MENU, SETTING

After Game Over

- The Game Over screen appears in the menu with RESTART and MENU
- MENU from the gameplay scene goes back through the Intro screen

Screen specifications

Screen	Elements	Behavior
Intro and menu	LOADING bar; START, SETTINGS, LEADERBOARD, QUIT	Intro loads for at least 2 seconds; START opens the name box; QUIT closes the app
Enter name	Input field, confirm button, cancel button	Confirm: saves the name, syncs it to the leaderboard, enters gameplay
HUD	SCORE, coins, distance (m), speed (m/s), power-up icons	Updated every frame; text changes only when the value changes
Pause	RESUME, RESTART, MENU, SETTING	Esc toggles; freezes all time; locked while waiting for a revive
Settings	Three sliders MASTER, BGM, VFX and a BACK button	In both the menu and the pause screen; dragging applies and saves immediately
Game Over	RESTART, MENU	Shown in the menu scene after skipping the revive

Seven audio files, three volume channels

Event	Sound	Notes
Throughout the run	Gameplay music	Loops, starts on entering the gameplay scene
Press jump	Jump sound	On key press, even while in the air
Press slide	Slide sound	On key press
Pick up a coin	Coin sound	On the coin pickup event
Stumble on an obstacle	Collision sound	Two variants
Lose	Lose sound (a fall)	Two variants

Three independent channels: master (default 100%), music (80%), effects (100%). The master channel adjusts the other two, and the levels are saved for next time. Picking up a power-up and reviving have no sound of their own.

Online score racing, no account needed

Anonymous sign-in

Through Unity Gaming Services as soon as the service starts; no account or password.

Choosing a name

- Pick a name before playing; if empty it becomes Runner_ plus a random number (1000 to 9999)
- Rename in the leaderboard
- Still playable offline

Submitting scores

- Only when the score is above 0
- On losing, and again when skipping the revive
- If offline it fails quietly, without affecting the run

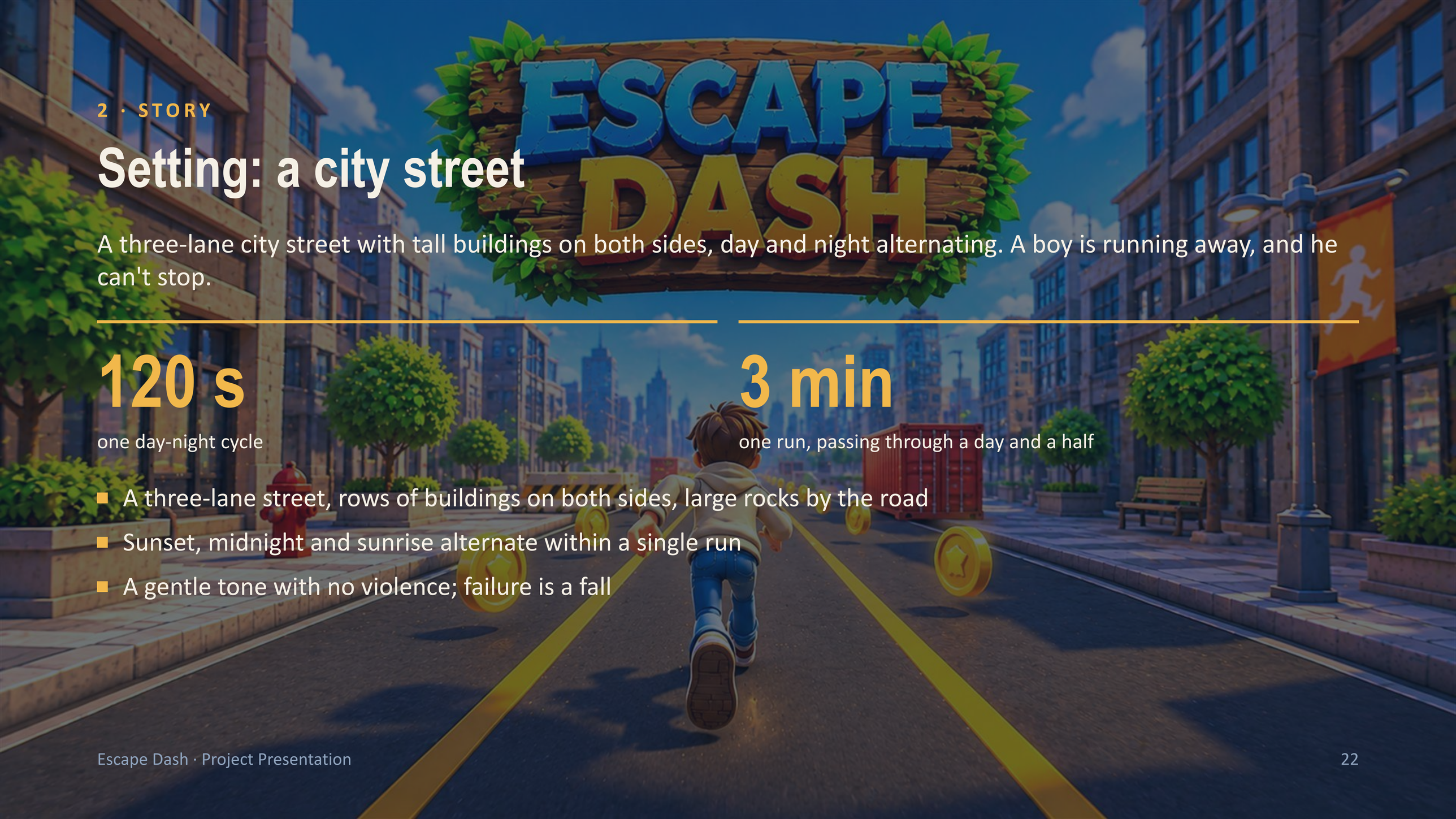
Leaderboard setup

ID High_Score; it must be set up on the Unity Cloud Dashboard to keep each player's highest score. The game does not compare scores itself.

The leaderboard screen

Element	Behavior
Open and close	Opens from the main menu; a near-black overlay locks the menu behind it; closes with the close button or by clicking the dark area
The list	Top 20: rank, name in capitals, score with thousands separators
Your own row	Light yellow background and a different name color
Refresh and rename	A refresh button; a name field and save button, locked while sending
Timeout	At most 2 seconds for real data, so the interface never freezes

Sample data: when the board opens, the game shows eight sample rows (9,920 down to 1,920), then replaces them with the real list if it loads within 2 seconds. If you are offline you only see the sample data, with no error message. This is the game's current behavior.



2 · STORY

Setting: a city street

A three-lane city street with tall buildings on both sides, day and night alternating. A boy is running away, and he can't stop.

120 s

one day-night cycle

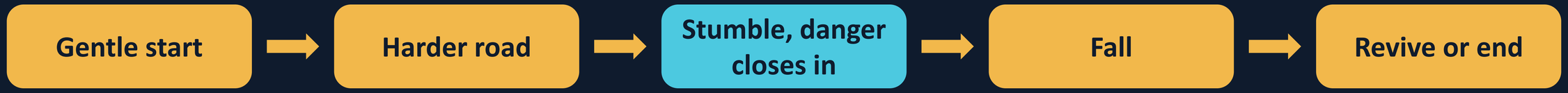
3 min

one run, passing through a day and a half

- A three-lane street, rows of buildings on both sides, large rocks by the road
- Sunset, midnight and sunrise alternate within a single run
- A gentle tone with no violence; failure is a fall

2 · STORY

How a run unfolds



The character

A boy running away from something, just like the game's title. No dialogue and no long story.

What chases

After the first stumble, danger closes in: you have 4 seconds not to stumble again, or you are caught.

Revive

Spend the coins you collected to get back up and keep running, keeping your score and distance.

Feelings along the road

Moment	Intended feeling	Design tool
Start	Relaxed, getting used to it	An empty road and a low speed of 3, rising to 6 within 6 seconds
Easy stretch (level 1)	Confident, rewarded	Only low barriers, 35% empty rows, plentiful coins, almost always a power-up
Medium stretch (levels 2, 3)	Focused	Overhead barriers added (level 2), then deadly obstacles (level 3)
Hard stretch (levels 4, 5)	Tense	All three types equally common (level 4), deadly obstacles most common (level 5), sparse coins

The table lists levels, not time order: the five levels appear in random order within each cycle of five segments.

Feelings on stumbling, losing and reviving

Moment	Intended feeling	Design tool
After a hard stretch	Relief	The pace eases when the shuffled bag lands on an easier level
When you stumble	A jolt	Stumble animation, collision sound, 4-second danger zone
When you lose	Regret, wanting to retry	Fall animation, the revive prompt
When you revive	A sigh of relief	Back to life in the middle lane on an empty road, speed reset

A 120-second day-night cycle



Sunrise Noon Sunset Midnight Sunrise

Position in cycle	Time of day	Sun color	Intensity
0%	Sunrise	Peach orange	0.4
25%	Noon	White	1.2
50%	Sunset	Deep orange	0.4
75%	Midnight	Dark blue	0

Starts at 15% of the cycle. The sun is fully off from about 58% to 92%, roughly 41 seconds of dark per cycle, so a three-minute run passes through a day and a half. Ambient light never drops below 0.45 and fog never below 0.15, so the road stays visible at night; shadows are angled 30 degrees.

Style and visual assets

Category	Description
Style	3D with colored models and materials
Environment	A road with lane markings, two city-strip patterns, many building models, large rocks by the road; a single theme
Character	A 3D human character with a full animation set
Items	Models of the coin, magnet bag, star and shoes; icons for the HUD
Obstacles	Concrete barriers, a type-3 barrier, a group of hazards
Interface	Menu and Game Over backgrounds, buttons, a dark overlay; a leaderboard art set (background, banner, crown, rank and score badges)

Shapes and colors must be easy to read so danger is recognized immediately.

Core technologies

Unity 6
Version 6000.3.9f1, Universal Render Pipeline 17.3, colored 3D graphics

C#
All gameplay logic: character, road, score, revive, interface

Cinemachine
A fixed camera behind the character, 60-degree field of view

Animator
An animation state machine: Run, JumpStart, JumpLoop, Roll, Hit, Death

TextMeshPro and uGUI
HUD, menu, pause, settings, revive panel, leaderboard

Unity Gaming Services
Authentication for anonymous sign-in and Leaderboards for the online ranking

Techniques in code

Object reuse
Road segments, obstacles, coins and power-ups are pre-built and reused; nothing is created during play

Shuffle bag
The five road levels are shuffled with Fisher-Yates, and a segment never repeats twice in a row

Shared events
Stumble, Game Over, coin pickup and revive pass through one event layer so systems depend on each other less

Storage
PlayerPrefs stores the record, coins, name and volume

Parameters outside the code
Spawn chances, power-up durations and the revive price are tuned in the Inspector and prefabs

Editor tools
Custom tools help build the HUD, build the low-poly character and check obstacle sizes

Unity packages and how much they're used

Package	Used for	Status
Cinemachine 3.1	The gameplay camera	In use
TextMeshPro	Text on every interface screen	In use
UGS Authentication, Leaderboards	Anonymous sign-in, the online leaderboard	In use
ProBuilder 6.0	Building graybox blocks for the early map	Used in the early phase
Input System 1.19	Installed; controls currently use the old Input	Installed
Timeline, AI Navigation	Installed in the project	Not used in gameplay

Technical foundation

Item	Details
Engine	Unity 6 (6000.3.9f1), Universal Render Pipeline 17.3
Language, input	C#; keyboard (Unity's old Input system)
Scenes in the build	IntroScence, MenuScence, GamePlay
Performance	Reuse of road segments, obstacles, coins and power-ups; HUD text updates only when the value changes
Storage	PlayerPrefs
Online services	Unity Gaming Services: Authentication (anonymous), Leaderboards (ID High_Score)
Main packages	Cinemachine, Input System, ProBuilder, Timeline, AI Navigation

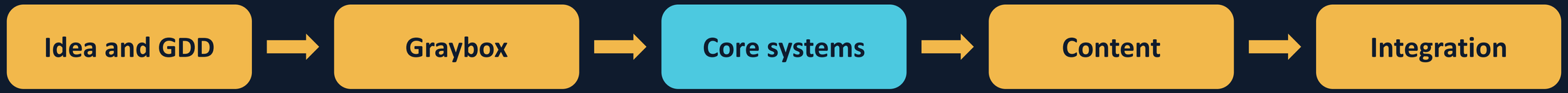
Event architecture

Event	Who raises it	Who reacts
Player stumbles (obstacle hit)	Obstacle	Character, audio
Game Over	Fatal obstacle, character (second stumble)	Character, road, score, power-ups, revive, interface, audio
Coin pickup	Coin	Score, audio
Revive	Revive	Character, road, score, interface

Systems communicate through one shared event layer so they depend less directly on each other. The menu and gameplay scenes synchronize through saved data: the Game Over flag, the last run's result, the name and the volume.

4 · PROCESS

How we worked



Idea and GDD

Pick a reference game, then write the loop, pillars and scope in the design document

Graybox and core systems

Build the rough block map, character controls, road spawner, obstacles, coins and score

Content and integration

Revive, leaderboard, power-ups, audio, day-night and interface; merge the branches and polish

Three phases

Phase	Dates	Main work
1. Foundation	30 Sep to 12 Oct	Graybox map, character controls, the endless road spawner, the random obstacle and coin spawner, score and record, main menu
2. Systems and content	16 Oct to 30 Oct	Revive, leaderboard, game lifecycle, audio, day-night cycle, a map with art, power-ups
3. Integration and polish	Early November	HUD, pause, Game Over, volume settings, the Intro screen, merging branches and fixing bugs

4 · PROCESS

How the team worked together

Work split by area

Each area has a deadline and a status on one shared task board: controls, road, items and score, interface and audio.

Source control with Git

Each member works on a separate branch, merging into a shared branch and the main branch.

Frequent integration

Branches were merged continuously from early November, and the whole loop was rechecked after each merge.

Cutting when needed

We dropped what didn't fit in time or didn't suit: the tall rocks, a follow camera, the old score system and the temporary runner.

Results after the phases

3
scenes in the build: Intro, Menu, GamePlay

5
difficulty levels

3
power-ups

1
online leaderboard

- Playable end to end, from the menu to Game Over and back again
- Enough secondary layers to bring players back: records, coins, revive, leaderboard

5 · THE PROBLEM

The problem we set

Build a complete 3D endless runner that is easy to pick up but worth replaying, in a short time and with a small team.

Goal

Learn the controls in seconds; understand why you lost every time; short runs that make you want to retry.

Players

Casual players who like games of a few minutes, no experience needed.

Constraints

About five weeks, a small team, PC keyboard, one road theme, one character.

Success criteria

Survive three minutes or reach 1,000 points in a single run.

5 · THE PROBLEM

Challenges and solutions

An endless road without hitches	Built from reusable segments, pre-created at load
Fair difficulty	Always a safe lane, five shuffled levels, enough time to react
Losing shouldn't discourage	The first stumble is only a penalty, a 4-second danger zone, revive with coins
A reason to replay	Score, record, a coin wallet and the online leaderboard
Readable	Three lanes, a fixed camera, nights never darker than you can see

Three outcomes of a collision

Head-on hit

- Lose instantly
- Even while invulnerable
- Hitting an obstacle straight on

Side graze

- Only a stumble, not a loss
- Happens when changing lanes right next to an obstacle
- Overlapping the obstacle by under 50% of the character's width

Top touch

- Only a stumble, not a loss
- Jumping over and landing on top of the obstacle
- Feet higher than the top surface (minus a tolerance of 0.35)

After a stumble: 0.8 seconds of invulnerability, then a 4-second danger zone. Stumbling again within the danger zone loses.

Three kinds of obstacle, three ways to dodge

Obstacle	How to dodge	Appears at
Low barrier	Jump over or change lane	Levels 1 to 5
Overhead barrier	Slide under or change lane	Levels 2 to 5
Deadly obstacle (a tall, long block)	Only by changing lane	Levels 3 to 5

All three share the direction-based collision rule: only a head-on hit loses instantly. The higher the level, the more tall and long obstacles there are, so changing lane at the right moment matters more.

An endless road built from segments

Parameter	Value
Length of each segment	20 units
Segments kept ahead	At least 3
Recycling	When a segment is 5 units behind the player
First segment	An empty segment starting at −10, not part of the bag
Shuffle bag	Five content segments shuffled (Fisher-Yates), dealt out then reshuffled, never the same segment twice in a row
Reuse	One copy created per frame at load, none created during play; if no spare copy is free, retry next frame

Four fairness guarantees

Always a way through

A continuous path across the rows: the safe lane shifts by at most one lane per row.

No impossible combinations

Each row has at most two obstacles across three lanes.

Enough time to react

7 units between rows: 2.3 seconds at speed 3 and 1.2 seconds at speed 6, while changing lane takes only 0.17 seconds.

Rewards are always reachable

Coins and power-ups always sit on the safe lane, so they can be taken without entering the danger zone.

Speed and pacing

3

starting speed (units per second)

+0.5

added per second

6

top speed, reached after 6 seconds

- A gentle start: every run begins with an empty segment and low speed
- After 6 seconds the road speed stays at 6; pacing relies on the five-level loop and does not rise over time
- The HUD has a speed gauge (m/s): it starts at 1.0, rises 0.1 per second, up to 20; a stumble drops it to 40%, then it recovers in 2 seconds. For now this is only a displayed number and does not control the road's scroll speed
- Tunable without changing the design: speeds, row spacing, spawn chances, danger-zone time, power-up durations, revive price

The score formula

Score = distance points + 10 × coins in the run

1
point per unit of road scrolled

10
points per coin

×2
with the Double Score power-up

- Distance comes from the total road scrolled, since the character stays still
- After reviving, score and distance are kept and keep adding up
- Coins picked up after losing do not count

Estimate for a perfect run

Time	Distance	Coins	Distance pts	Coin pts	Total
30 s	171	48	171	480	651
60 s	351	98	351	980	1,331
120 s	711	198	711	1,980	2,691
180 s	1,071	299	1,071	2,990	4,061

Collecting every coin with no power-ups: about 28 coins per 100 units of road. Coins make up about 74% of the score, and the 1,000-point mark comes after about 44 seconds.

Coin economy and saved data

Coin sources

- Collecting coins on the road: +1 to the wallet
- Doubled while Double Score is active

Coin sinks

- Revive: −50 coins each time, with no limit
- The only place coins are spent
- One revive equals about 180 units of road, about 30 seconds at speed 6

Saved on the device

- Highest score, longest distance, total coins
- Player name, volume
- Records are written only on losing, once per run

What happens when you lose

1. A losing collision: the road stops scrolling, the fall animation plays, scoring stops, power-ups are cleared and the lose sound plays
2. The game records the best score (if any), saves the run's result and submits the score to the leaderboard (if above 0)
3. After 0.3 seconds and once the fall animation finishes (waiting at most 3 seconds), a dark overlay and the revive panel appear, with a real-time 5-second countdown
4. The player chooses to revive, skip, or lets the time run out (same as skipping)

Skip or timeout: the score is submitted, the Game Over flag is set and the menu scene loads, where the Game Over screen with RESTART and MENU appears.

Coins turn into a breath

50

coins per revive

5 s

countdown to decide

≤ 3 s

wait for the fall animation to finish before the panel shows

- Enough coins: spend 50 coins and come back in the middle lane on an empty road
- Not enough coins: the button is dimmed, and Space or R does nothing
- Skip or timeout: the run ends
- No limit on revives; only the wallet limits you

What revive resets and what it keeps

Part	Reset	Kept
Character	Back to the middle lane at ground level; clears the stumble counter, danger zone, invulnerability and slide; Run animation	—
Road	Recycles every segment, places a safe empty segment, speed back to 3, keeps spawning ahead	—
Score, distance, this run's coins	—	Kept and keep adding up
Coin wallet	−50	The rest is kept
Power-ups	Already cleared on losing	Not given back

Text and colors on the revive panel

Element	Text shown	Color
Title	DO YOU WANT TO REVIVE?	—
Price	Cost: 50 Coin	Yellow
Balance (enough)	You have: N Coin (Enough)	Green
Balance (short)	You have: N Coin (Need M more)	Red
Countdown	Auto skip in: Ns	Orange
Buttons	REVIVE (-50 Coin) [Space]; if short: NOT ENOUGH COINS (Need 50), button dimmed; SKIP [Esc]	—

The text on the revive panel is already in English; the name entry, the leaderboard and the fallback Game Over panel are still in Vietnamese. Esc here is only for skipping and does not pause the game.

5 · THE PROBLEM

The answer: a short loop that's easy to replay

1 · Observe
Look at the row of obstacles and find the empty lane

2 · Choose
Change lane, jump or slide

3 · Dodge and collect
Dodge obstacles, collect coins and power-ups

4 · Score
Points, distance and coins rise instantly

5 · Lose, replay
Revive with coins or replay right away

Easy to read. Losing makes you want to try again.

APPENDIX

Balance parameters (1)

Group	Parameter	Value
Road speed	Start / max / acceleration	3 / 6 / 0.5 per second
Road	Segment length / segments kept ahead / recycle after	20 / 3 / 5 units
Rows	First row position / spacing / end margin	3 / 7 / 2
Character	Lane spacing / lateral speed	2 / 12
Jump	Force / gravity / Super Jump multiplier	8 / −22 / 1.8
Slide	Duration / remaining height	1.1 s / 50%
Stumble	Danger zone / invulnerability / animation	4 / 0.8 / 0.6 s
Coin	Value / coins per row / spacing / buffer	1 / 3 / 1.5 / 24

APPENDIX

Balance parameters (2)

Group	Parameter	Value
Score	Per unit of road / per coin / doubling factor	1 / 10 / 2
Power-up	Magnet / Double Score / Super Jump (seconds)	10 / 10 / 8
Magnet	Radius / pull speed / pickup distance	8 / 28 / 0.4
Power-up spawning	Minimum between two power-ups / type weights	4 segments / 2:2:2
Revive	Price / countdown / max wait for the fall animation	50 coins / 5 s / 3 s
Day-night	Cycle length / starting point	120 s / 15% of the cycle
Leaderboard	Rows shown / wait for real data	20 / 2 s
HUD speed	Start / gain per second / max / left after a stumble	1 / 0.1 / 20 / 40%

WRAP-UP

Thank you for listening

3	4	5	3
lanes	actions	difficulty levels	power-ups

- Simple to start, hard to master, and always gives the player a reason to try again
- Questions are welcome